RAKSHITHA K

Student

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ABOUT ME

Motivated B.Tech (Hons.) student at RV University with a strong passion for AI, robotics, automation, and research. Enthusiastic about solving real-world challenges using machine learning and emerging technologies. Currently building hands-on expertise in Java, SQL, Python, C, and OpenPLC, with project experience in operating systems, databases, and environmental monitoring. Deeply interested in machine learning and deep learning, and eager to apply and expand my skills while contributing to impactful and innovative solutions.

EDUCATION

2023 - 2027
RV University

BTech (H) Minors in Robotics and IA

I have completed my 2nd year (4th semester) of B.Tech in engineering and have a strong passion for learning new things and expanding my knowledge.
CGPA till 5th SEM: **8.94**

2021 - 2023
RV PU College

Pre-University Education (Class 11 & 12)

Secured **90%** in 12th Board Exams (PUC)

2010 - 2020
N.E.T Public school

Schooling (Class 1 – 10)

Secured **91%** in 10th Board Exams (**ICSE**)

INTERNSHIPS AND PROJECTS

Jan 2025 – May 2025
Agile Software Engineering Project

AI-Powered Drug Discovery & Quantum Protein Folding

Built a Streamlit-based app using ML, DL, and Quantum Computing (QML/QNN) to automate drug discovery and protein folding. Deployed with Docker & Kubernetes, integrated CI/CD pipelines, and visualized molecular properties to accelerate pharmaceutical research.

Feb 2025 – Apr 2025
Smart Cities (IIOT)Research Paper Ongoing

Smart Emergency Routing using Deep Reinforcement Learning

Designed an actor-critic DRL model to dynamically route emergency vehicles and control traffic signals in a simulated smart city grid, achieving faster response times through real-time decision-making and green corridor creation.

Jan 2025 – Mar 2025
NLP & ML Project , Research

MailOrg – Intelligent Email Clustering and Summarization

Built a smart email management system using BERT embeddings, K-Means clustering, and BART summarization to group semantically similar emails and auto-generate concise summaries. Achieved effective inbox organization across 12,000+ real emails, enabling users to delete, archive, or prioritize messages efficiently.

June 2025 – July 2025
External Internship in Zodopt Technologies

SalesBUDDY - AI-Powered B2B Sales Automation System

Developing an AI-powered sales assistant to automate lead profiling, BANT scoring, personalized outreach, and response tracking.
Building front-end forms, CSV uploads, and CRUD APIs using React and FastAPI; integrated automation tools like LangGraph and n8n for lead analysis.

June 2025 – July 2025

Research and Build, Ongoing

Active Learning for Multimodal Models

Experimenting with active learning on CIFAR-10 using VGG16 and advanced strategies like CoreSet and VAAL.
Evaluating sampling efficiency for model improvement with limited labels.
Plan to generalize optimal strategy to different model architectures and apply it in multimodal learning setups.

Oct 2025 – Nov 2025

Research and Build, Ongoing

Attention-Based Parkinson's Disease Detection from Speech

Built an explainable deep learning framework for early Parkinson's detection using sustained vowel speech, combining Mel-spectrograms with VGG16, BiLSTM-Attention, and mLSTM architectures. Integrated Monte Carlo Dropout for uncertainty estimation and Grad-CAM/EigenCAM for interpretability, achieving ~80% accuracy with ROC-AUC ≈ 0.87.

June 2024 - July 2024

Machine Learning (Cybersecurity) at Dhee Centre for AI and Data Science

Anomaly Detection in Network Traffic

Developed a cybersecurity project focused on network anomaly detection, leveraging the Random Forest algorithm to accurately identify unusual traffic patterns and potential security threats, thereby enhancing network intrusion detection and response capabilities.

Sept 2025 – Oct 2025

IOT project (Both Hardware and Software)

Smart Classroom Light and Temperature Control

Designed an IoT- and ML-based smart classroom system using ESP32 to automate lighting, temperature, ventilation, and safety control. Integrated sensors (LDR, DHT22, MQ-135, MQ-2, flame) with predictive ML models for air quality, fire risk, and ventilation intensity, enabling real-time autonomous actuation of fans, lights, windows, and alarms. Achieved improved energy efficiency, indoor comfort, and rapid safety response through sensor-driven and data-guided control.

SKILLS

- **Languages:** Python, Java, C++, C, SQL
- **Libraries/Frameworks:** TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, PennyLane, Librosa, Keras
- **Data & Cloud Tools:** PySpark (learning), AWS (EC2, S3), Docker, Kubernetes, MongoDB, Git, Jira, ESP32 (Embedded C++), MQTT / HTTP
- **Domains:** Machine Learning, Deep Learning, Cloud Computing, Data Engineering, Smart City Systems, Quantum ML, DevOps, Agile (Scrum), Active Learning, Explainable AI (XAI) ,IoT & Embedded Systems
- **Soft Skills:** Problem-Solving, Communication, Teamwork, Time Management, Adaptability